

Decision Tree Regression

Dataset

→ Bxp

→ 2

→ 2.5

3

4

4.5

Gap

yes

yes

no

no

yes

O/P

Salary

40K

42K

52K

60K

56K

50K

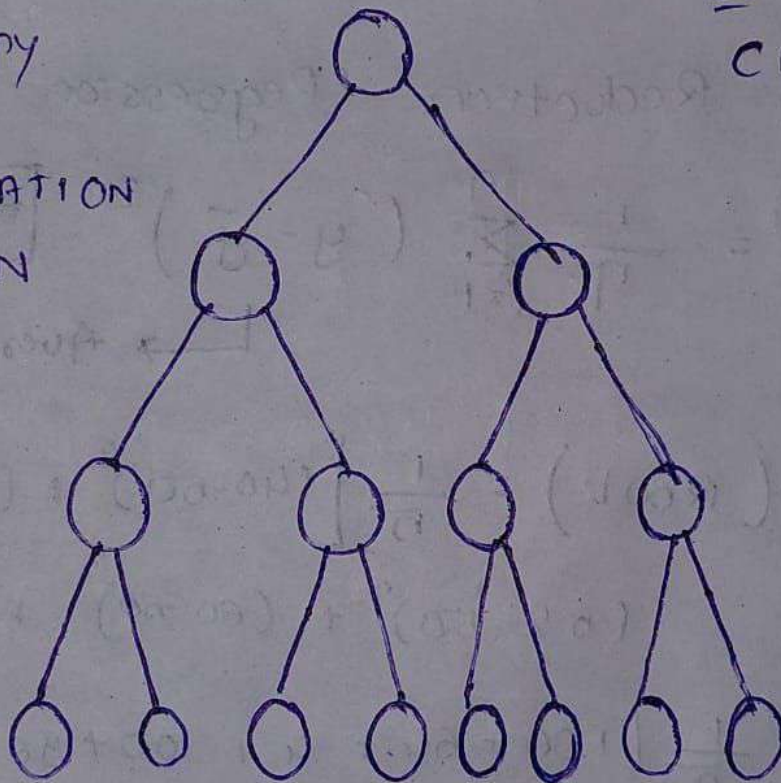
Entropy

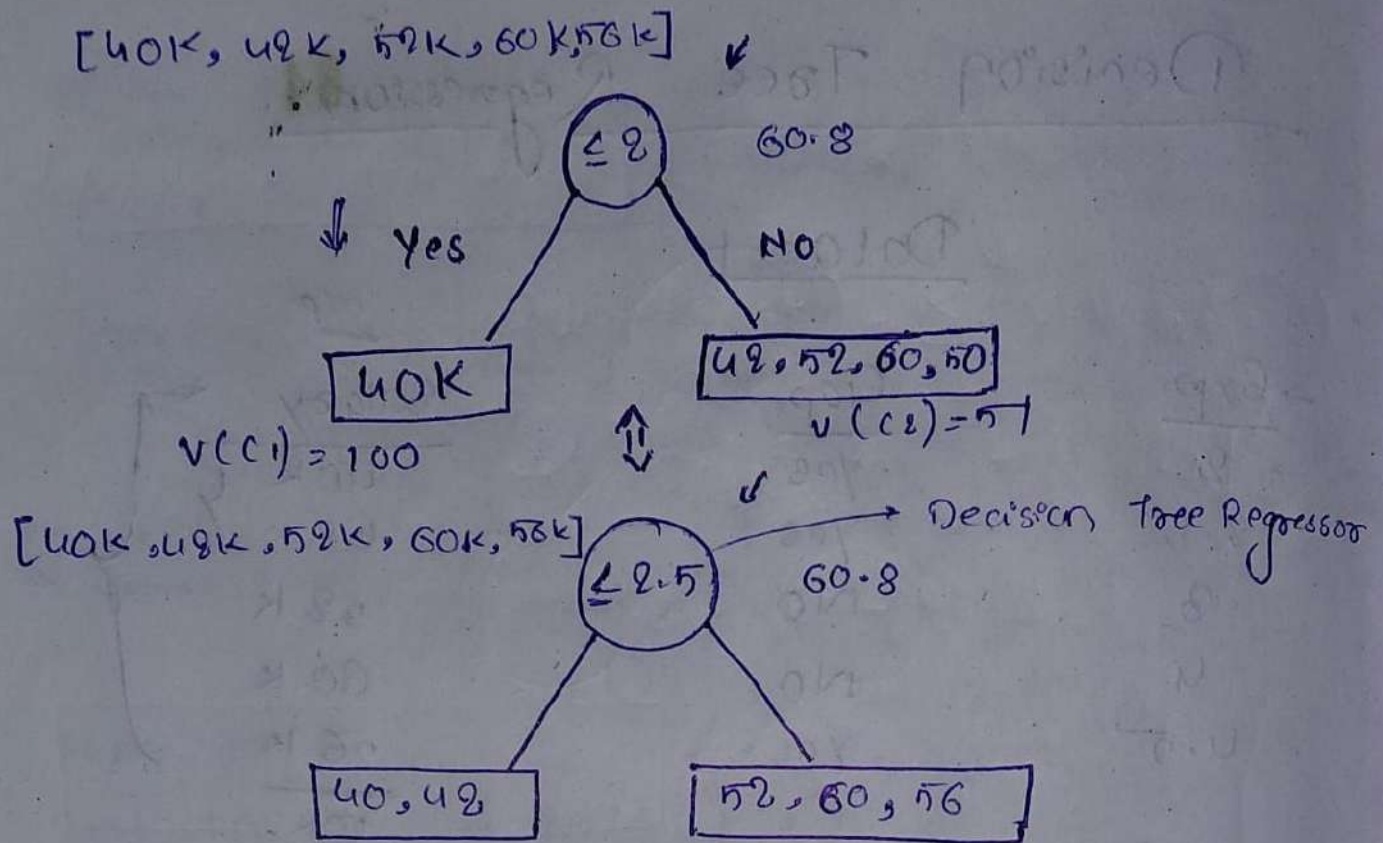
Gini

INFORMATION

GAIN

CART





Variance Reduction (Regression Problem)

① Variance = $\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2$ Mean Squared Error

Average

Variance (Root) = $\frac{1}{5} [(40 - 50)^2 + (42 - 50)^2 + (52 - 50)^2 + (60 - 50)^2 + (56 - 50)^2]$

= $\frac{1}{5} [100 + 64 + 4 + 100 + 36]$

= 60.8

$$\text{Variance } (C_1) = \frac{1}{n} \sum_{i=1}^n (y - \bar{y})^2$$

85

$$= \frac{1}{1} (40 - 50)^2$$

$$= 100$$

$$\text{Variance } (C_2) = \frac{1}{n} \sum_{i=1}^n (y - \bar{y})^2$$

$$= \frac{1}{4} [(42 - 50)^2 + (50 - 50)^2 + (60 - 50)^2 + (56 - 50)^2]$$

$$= \frac{1}{4} [64 + 0 + 100 + 36]$$

$$= 51$$

Variance Reduction

$$= \text{Var}(\text{Root}) - \sum \omega_i^2 \text{Var}(\text{Child})$$

$$= 60.8 - \left[\frac{1}{5} \times 100 + \frac{4}{5} \times 51 \right]$$

$$= 60.8 - 20 - 40.8$$

$\text{Variance Reduction} = 0$

$$\text{Var}(C_1) = \frac{1}{2} \left[(40 - 50)^2 + (42 - 50)^2 \right]$$

$$= \frac{1}{2} [100 + 64]$$

$$= \frac{164}{2}$$

$$= 82$$

85

$$\text{Var (child)} = \frac{1}{3} [(52-50)^2 + (60-50)^2 + (56-50)^2]$$

$$= \frac{1}{3} [4 + 100 + 36]$$

$$= \frac{140}{3} = 46.66$$

Variance Reduction

$$= 60.8 - \left[\frac{2}{5} \times 82 + \frac{3}{5} \times 46.66 \right]$$

$$= 60.8 - [82.8 + 27.996]$$

$$= \underline{\underline{+0.004}}$$

0

0.004

Variance Reduction (Left split) $<$ VR (Right split)

left tree

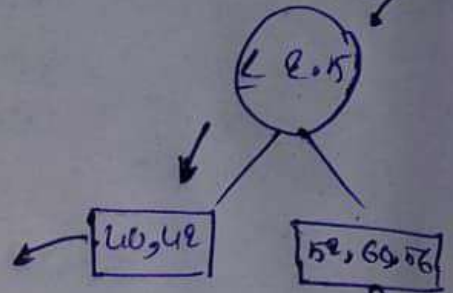


Leaf Node

$$\text{O/P} = \frac{40+42}{2}$$

$$= 41$$

Test data



$$\frac{52+60+56}{3}$$

$$= 56$$

O/P

Decision Tree Regressor

Note: The average output is a output of a decision tree regression.